Fairness, Bias, and Interpretability

Ask what a model does, why, and to whom

Broader Context · Foundations of Machine Learning

Bias Can Enter at Every Stage

Data bias — Historical inequity or underrepresentation in *who was sampled* becomes the pattern the model learns to reproduce.

Label bias — The target variable y is often a proxy for what we actually want (e.g., "arrested" as a proxy for "committed a crime"), and past human decisions can encode discrimination directly into that proxy.

Measurement / proxy bias — Features that look neutral (zip code, school name) can correlate strongly with a protected attribute like race or income, letting the model discriminate indirectly even when that attribute is excluded from the input.

1. Data
2. Labels
3. Features
4. Model decision

"Fairness through unawareness" — simply deleting the protected attribute column — fails precisely because bias can re-enter at stage 3 through correlated proxies.

Case Study — COMPAS Recidivism Risk Scores

The system

COMPAS is a proprietary tool used by U.S. courts to score a defendant's risk of reoffending, informing bail and sentencing decisions.

ProPublica's 2016 audit

Black defendants who did *not* reoffend were nearly twice as likely as white defendants to be labeled high-risk; white defendants who *did* reoffend were more often mislabeled low-risk.

Northpointe's rebuttal

The tool's vendor countered that COMPAS was well-calibrated: within each risk score bucket, the actual reoffense rate was similar across race.

Both sides were right

This became the canonical demonstration that calibration and error-rate balance are different, formally incompatible fairness properties.

Three Group Fairness Definitions

Demographic parity

$P(\hat{Y}{=}1 \mid A{=}a) = P(\hat{Y}{=}1 \mid A{=}b)$ — equal positive-prediction rates across groups a, b, regardless of the true label.

Equalized odds

Equal true-positive rate *and* equal false-positive rate across groups, conditioned on the true label Y.

Predictive parity (calibration)

$P(Y{=}1 \mid \hat{Y}{=}1, A{=}a) = P(Y{=}1 \mid \hat{Y}{=}1, A{=}b)$ — equal precision: a positive prediction means the same thing in every group.

Audit method

Compute the same confusion-matrix-derived metrics from model evaluation (TPR, FPR, precision) separately within each protected group, then compare.

Fairness Metrics Can Be Mutually Incompatible

The impossibility result — Chouldechova (2017) and Kleinberg, Mullainathan & Raghavan (2016) proved: if group base rates $P(Y{=}1\mid A{=}a) \ne P(Y{=}1\mid A{=}b)$ and the predictor is imperfect, calibration and equalized odds cannot both hold exactly.

Not an engineering bug — No better optimizer, more data, or bigger model removes this mathematical conflict; it follows from the definitions themselves.

A value judgment, not a math problem — The application and the people affected must determine which kind of error the system should be more careful to avoid.

$$\begin{aligned} P(Y{=}1 \mid A{=}a) &\neq P(Y{=}1 \mid A{=}b) \\ \text{and imperfect prediction} &\implies \text{fairness tradeoffs} \end{aligned}$$

This is precisely the COMPAS situation: reoffense base rates differed by race in the data, so equal calibration (Northpointe's claim) and equal false-positive rates (ProPublica's claim) could not both hold.

Individual and Group Fairness Can Conflict

Individual fairness

Dwork et al. (2012): "similar individuals should be treated similarly" — formally, a Lipschitz condition $d(\hat{Y}(x_1), \hat{Y}(x_2)) \le L \cdot d(x_1, x_2)$ on some similarity metric d.

The hard question

Defining "similar" requires a task-specific, value-laden distance metric — who decides which features count, and how much each one weighs?

The tension

Enforcing a group-level statistic (e.g., demographic parity) can require treating two individuals who are nearly identical except for group membership *differently*.

Group fairness asks "do the group-level statistics match?" Individual fairness asks "are similar people treated similarly?" A system can satisfy one while visibly violating the other — there is no metric that automatically reconciles both.

Interpretability Has Global and Local Forms

Global

Understand overall model behavior across all inputs: linear-model coefficients, a small decision tree's splits, or aggregated feature importance from a random forest.

Local

Explain one specific prediction, even when the whole model (a deep network, a large ensemble) is too complex to summarize in one description.

Why it matters

Debugging unexpected behavior, building user and regulator trust, meeting legal compliance (e.g., adverse-action notices in lending), and running the fairness audits from earlier in this deck.

Complexity–interpretability spectrum across this course: linear/logistic regression (fully global, coefficients are the explanation) → single decision tree (global, trace root to leaf) → random forest / gradient boosting (global importance only via aggregation) → neural network (neither is naturally readable — needs dedicated local tools like SHAP).

SHAP Explains One Prediction Additively

Baseline — Start at $\mathbb{E}[f(X)]$, the model's average prediction over the whole training set.

Feature contributions — Each SHAP value ϕ_i is the feature's average marginal contribution across every possible ordering of features, borrowed from Shapley values in cooperative game theory.

Additive guarantee — Contributions sum exactly to this case's prediction: $f(x) = \mathbb{E}[f(X)] + \sum_i \phi_i$.

Worked example: a loan-default model, starting from the 30% average default rate.

Fairness and Interpretability

Audit sources

Bias enters via data, labels, and proxy features — check all three, not just the model.

Choose values

Group fairness definitions provably conflict when base rates differ — pick deliberately, don't default.

Individual lens

Also ask whether similar people are treated similarly, not only whether group statistics match.

Explain

Use global tools (coefficients, feature importance) and local tools (SHAP) together.

Next: where classical ML ends and modern deep learning begins.